

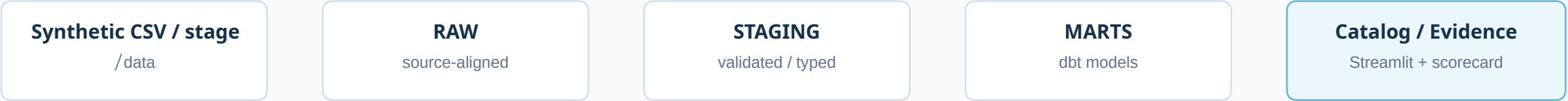
What Snowflake Changes Next

A practical roadmap for OUR banking governance project

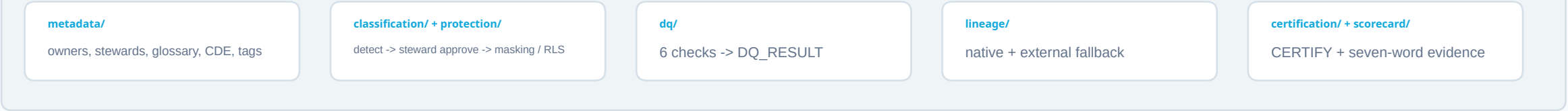
**Keep RAW -> STAGING -> MARTS and the bank-specific trust controls.
Use newer Snowflake capabilities to remove manual metadata, lineage and deployment work - without rewriting the pipeline.**

1. Project overview & current architecture

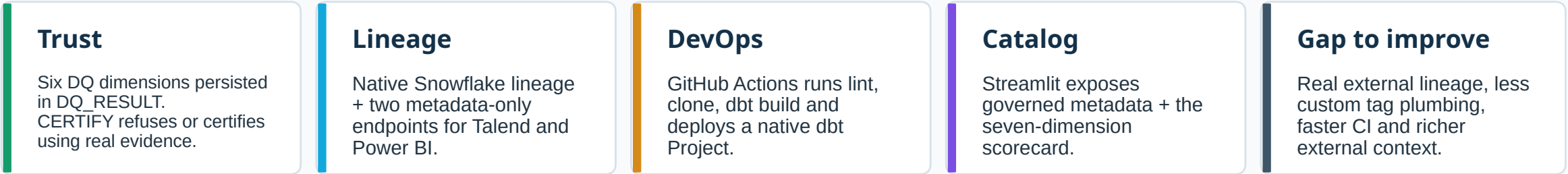
DATA FLOW TODAY



Governance control plane



WHAT IS ALREADY REAL IN MAIN



2. Important Snowflake developments for the next 12 months

Feature	Status	Why it matters to this project	Our posture
External Lineage	GA	OpenLineage-based external + column lineage; replaces metadata-only fallback.	ADOPT NOW
DCM Projects	GA	Declarative Snowflake objects as code with plan/deploy across environments.	ADOPT SELECTIVELY
dbt Slim CI / defer	GA	Build/test only modified nodes + downstream; reuse production artifacts.	ADOPT NOW
Snowflake-provided tags	PUBLIC PREVIEW	Standard CERTIFICATION_STATUS / SENSITIVITY vocabulary.	PILOT
Tag-based protection policies	PUBLIC PREVIEW	Tag-driven row access, aggregation, projection and join policy attachment.	PILOT
AI mode for classification	PUBLIC PREVIEW	LLM-assisted semantic detection can enrich the current STAGING classification profile.	PILOT
Horizon Context connectors	*PRIVATE PREVIEW	Bring external DB / Power BI / dbt context into Horizon Catalog.	MONITOR / PILOT
Snowflake-managed MCP	GA	Optional governed, read-only AI tool access to catalog context later.	OPTIONAL LATER

3. External Lineage - the first change I would make

GA

TODAY

Talend + Power BI are represented as two metadata-only rows around native Snowflake lineage.

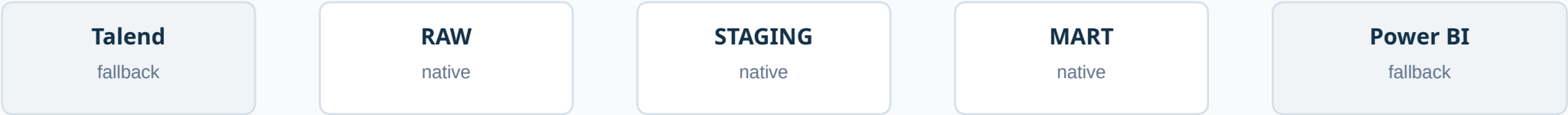
EXTERNAL LINEAGE / OPENLINEAGE

External tools send COMPLETE OpenLineage events to Snowflake; object and column edges join the native graph.

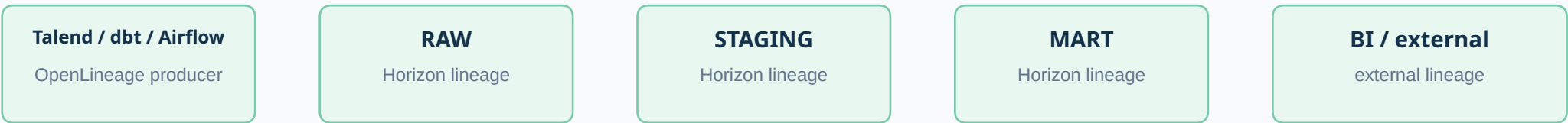
AFTER

Real external nodes become queryable in GET_LINEAGE / Snowsight and usable for impact analysis.

CURRENT ARCHITECTURE



IMPROVED ARCHITECTURE



PROJECT CHANGE

Retire lineage/03_seed_external_lineage_fallback.sql as the primary mechanism; keep it only as a fallback.

EFFORT

Medium: configure producers, endpoint auth and INGEST LINEAGE privileges.

VALUE

Traceability + impact analysis + less manual metadata maintenance.

4. Classification & protection - more automation, same control gates

AI-assisted classification

WHAT IT IS

Public Preview. A classification profile can use an LLM to identify additional semantic categories beyond standard detection.

HOW IT FITS

Enable it on the existing STAGING classification profile in DEV and compare precision/recall to sealed_sensitive_columns.json.

WHAT STAYS

The review queue remains. Only steward-confirmed results set the governance CLASSIFICATION tag.

Provided governance tags

WHAT IT IS

Out-of-the-box tags in SNOWFLAKE.TAGS, including CERTIFICATION_STATUS and SENSITIVITY.

HOW IT FITS

Map equivalent custom concepts where values align. Keep DATA_OWNER, DATA_STEWARD, DOMAIN, LAYER, CDE and SOURCE_SYSTEM custom.

BENEFIT

Less taxonomy drift and a vocabulary Snowflake services can interpret consistently.

Tag-driven protection

WHAT IT IS

Masking by tag is GA; row access, aggregation, projection and join policy attachment via tags are Public Preview.

HOW IT FITS

Keep current masking pattern. Pilot tag-based row access so new protected objects inherit control with less object-specific SQL.

BANKING EXTENSION

Data Movement Policies (GA) can attach to tags and alert/block exports, UI downloads, programmatic fetch or agent access.

5. Horizon Context - governed external business context

WHAT SNOWFLAKE IS INTRODUCING

Metadata connectors

Collect schemas, query logs, dashboard definitions and pipeline metadata from external systems such as Power BI, dbt, PostgreSQL and SQL Server.

Connected context

Combine technical metadata, usage, lineage, descriptions and semantics so humans and AI can discover authoritative assets.

Select Star technology direction

Select Star technology is being folded into Horizon to broaden external metadata, lineage and discovery. Treat this as a direction, not a separate dated integration promise.

WHAT THIS MEANS FOR OUR PROJECT

CONNECTORS: PRIVATE PREVIEW

TODAY

metadata/ harvests Snowflake metadata; lineage/ stores external fallback edges; catalog_app/ presents our governed evidence store.

WITH CONNECTORS

Horizon can collect Power BI / dbt / external DB context, reducing duplicate hand-maintained external metadata and improving end-to-end discovery.

OUR CHANGE

Keep GOVERNANCE.CATALOG as the audit/control evidence layer. Enrich the Streamlit view from Horizon context instead of rebuilding every external relationship ourselves.

AI-generated descriptions can help draft missing documentation - still route drafts through approval.

No committed GA date for metadata connectors: monitor availability and compare coverage before removing custom metadata.

6. Development & operations - easier to change, cheaper to test

DCM Projects

GA

TODAY

Many ordered SQL files are executed with Snowflake CLI. Reproducible, but dependency ordering and environment drift become harder as the repository grows.

AFTER

Describe supported Snowflake objects as desired state -> preview a plan -> deploy idempotently across DEV / TEST / PROD.

WHERE IT FITS

setup/ + selected metadata/ and governance objects. Refactor gradually; do not rewrite everything. Some governance definitions/attachments in DCM are still Preview.

VALUE Safer deploys

EFFORT Medium

dbt Projects: Slim CI + defer

GA

TODAY

The current dbt_ci workflow creates a zero-copy clone and runs a full dbt build even when a pull request changes only a small part of the DAG.

AFTER

Import recent production artifacts and run state:modified+ with --defer so CI validates changed nodes + downstream dependencies while reusing unchanged upstream production relations.

WHERE IT FITS

dbt/ + GitHub Actions. Keep zero-copy isolation for writes, but stop paying for unnecessary full-DAG execution on every change.

VALUE Less CI compute

VALUE Faster feedback

7. Additional banking controls worth mentioning - but label them correctly

Storage Lifecycle Policies

GA

WHAT IT REALLY IS

Schema-level policies that can automatically archive or expire table rows based on defined conditions. Useful for retention, deletion and storage-cost controls.

HOW IT HELPS US

For a real bank, this can implement record-lifecycle rules after legal/compliance defines them. It is an optional extension beyond the take-home.

Data Movement Policies

GA

WHAT IT IS

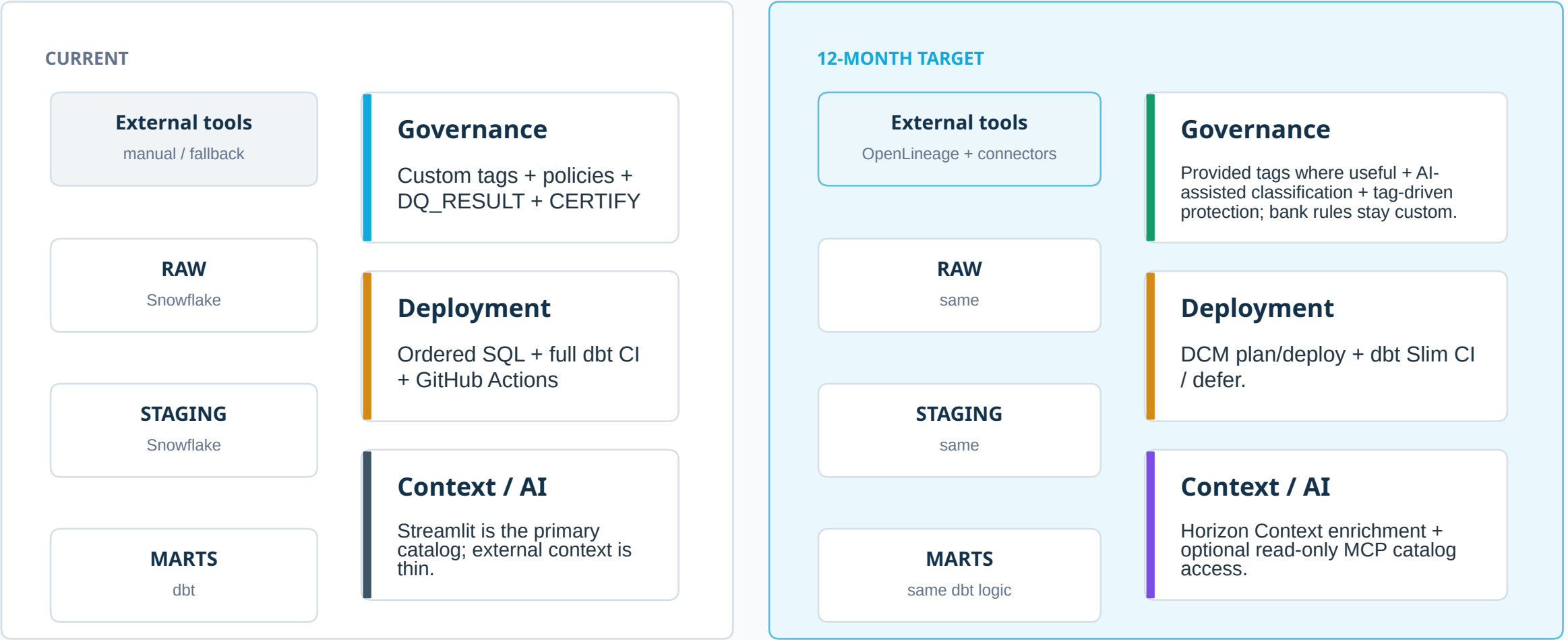
Tag-attachable controls that can allow, alert on, limit or block sensitive data movement such as stage export, UI download, programmatic fetch and agent access.

HOW IT HELPS US

Once our confirmed CLASSIFICATION/SENSITIVITY tags identify restricted assets, DMP adds exfiltration guardrails without changing the consumer table design.

8. Current -> 12-month target architecture

The data layers stay stable; metadata, lineage, CI/CD and protection become more Snowflake-native.



.Architectural conclusion: evolution, not rewrite. The pipeline remains RAW -> STAGING -> MARTS; the control plane becomes less manual and more portable

9. 12-month adoption roadmap & decision gates

NOW -> 3M

Adopt

External Lineage PoC -> real external edges
Slim CI / defer
Selected low-risk DCM objects

3 -> 6M

Pilot

Snowflake-provided tags in DEV
AI mode classification against sealed list
Tag-based row-access + Data Movement patterns in DEV

6 -> 9M

Monitor / pilot

Horizon Context connectors if available
Compare Power BI / dbt context with manual governance store
Measure metadata freshness / coverage

9 -> 12M

Decide

Read-only MCP catalog assistant only if useful
External catalog gate review
Decide whether custom fallback metadata can be retired

EXTERNAL CATALOG GATE

Add one only if...

- Critical non-Snowflake systems remain invisible
- Formal stewardship / attestation workflow is still missing
- Regulators need cross-platform evidence Horizon cannot provide
- An enterprise catalog is already adopted and valuable

TOP 3 RISKS

Control the adoption risk

1. Preview dependency -> keep mandatory controls on GA paths
2. Metadata staleness -> harvest, reconcile, recertify
3. Wider connector / agent surface -> least privilege + masking/RLS + audit

FINAL RECOMMENDATION

What I would do

ADOPT External Lineage, Slim CI, selected DCM
PILOT Provided tags, AI classification, tag-based policies / DMP
MONITOR Horizon Context connectors
LATER MCP when a real read-only catalog use case appears

Keep CERTIFY, DQ, RESULT, steward approval and ownership rules - Snowflake simplifies the plumbing; the bank still defines trust